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Mathematically, the consumer's surplus (CS) can be defined as

CS • (Area under the demand curve from x • 0 to x • x_0) • (Area of the rectangle OX_0BP_0)

$$CS • [\int_0^{x0} p(x)dx] • x_0p_0$$

Example:12.28

If the demand function is P • 35 • 2x • x^2 and the demand x_0 is 3, what will be the consumer's surplus?

Solution

Given demand function,

$$P • 35 • 2x • x^2$$

for x • 3

$$P • 35 • 2(3) • 3^2$$

$$• 35 • 6 • 9$$

$$P • 20$$

Therefore,

CS • (Area of the curve below the demand curve from 0 to 3) • Area of the rectangle (20 x 3 • 60)

$$CS • \int_0^3 (35• 2x• x^2)dx• (20×3)$$

$$• \left[35x - 2\frac{x^2}{2} - \frac{x^3}{3}\right]_0^3 • 60$$

$$• 35(3) • 2(\frac{3^2}{2}) • \frac{3^3}{3} • 60$$

$$• 105 • 9 • 9 • 60$$

$$• 27 \text{ Units.}$$

12.5.6 Producer's surplus

Example 12.29

Given the demand function P_d • 25 • Q^2 and the supply function P_s • 2Q • 1. Assuming pure competition, find (a) consumers surplus and (b) producers surplus. (P_d • Demand price; P_s • Supply price)

Solution:

For market equilibrium, $P_d = P_s$

$$25• Q^2 • •2Q• 1$$

$$0 • • 25 • •Q^2 • •2Q• 1$$

$$0 • •• 24 • •Q^2 • •2Q$$

$$Q^2 • •2Q • 24 • •0$$

$$Q^2 • •6Q • 4Q • 24 • •0$$

$$Q(Q • •6) • 4(Q • •6) • •0$$

$$(Q • •6)(Q• 4) • •0$$

So, Q • •4 or Q • •• 6. Since Q cannot be equal to • 6,

$$Q • 4$$

When Q• 4, P_d • 25 • 4^2• 9;

$$P_s • 2(4) • 1• 9$$

Consumers' surplus • $\int_0^4 (25 • Q^2)dQ •$ (9 X 4)

$$• \left[25Q - \frac{Q^3}{3}\right]_0^4 - 36$$

$$• [(25)(4) • \frac{1}{3} (4)^3] • (0) • 36$$

$$• [100 • \frac{64}{3}] • (0) • 36 • 42.67$$

Producers' surplus P_s

$$(P_{s)} • (9×4) • \int_0^4 (2Q• 1)dQ$$

$$• 36 - \left[(Q^2+Q)\right]_0^4$$

$$• 36 • (16 • 4) • 16$$